

Unit 3 Guided Notes — Equations & Inequalities

Keep it balanced. Undo step by step. Flip the sign when you divide by a negative.

Standards: NC.8.EE.7

 **Blank Worksheet (PDF)**

 **Answer Key (PDF)**

 **Official PDF Packet**

 **EC/Modified Notes (PDF)**

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. Work the blanks on your printed copy first, then click any blank, step, or check button on screen to reveal the answer.

1 One-Step Equations / Inverse Operations

✦ **THE BIG IDEA:** INVERSE OPERATIONS are used to isolate the variable. Whatever you do to one side, you MUST do to the other to keep the equation balanced!

Operation	Inverse	Example
Addition	Subtraction	$x + 5 = 12$
Subtraction	Addition	$x - 3 = 8$
Multiplication	Division	$3x = 15$
Division	Multiplication	$x/4 = 2$

Match the Operation to its Inverse

MATCH Click an operation, then its inverse.

Operation

Addition

Subtraction

Multiplication

Division

Inverse

Division

Subtraction

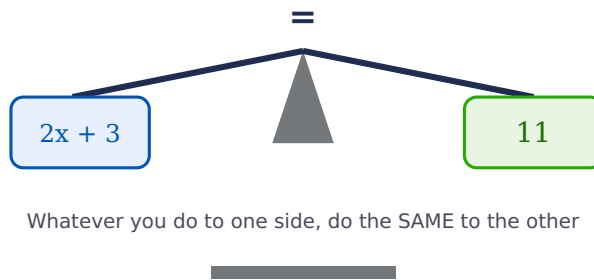
Multiplication

Addition

0 of 4 matched

2 Two-Step Equations

✦ **TWO STEPS:** Step 1 — Undo addition/subtraction. Step 2 — Undo multiplication/division.



Example 1: $-2x + 4 = 8$

Step 1 – Undo addition (subtract 4): $-2x + 4 - 4 = 8 - 4 \rightarrow -2x = 4$

✓ Step 2 – Undo multiplication ($\div -2$): $x = -2$

Example 2: $w/2 + 6 = 7$

Step 1 – Undo addition (subtract 6): $w/2 = 1$

✓ Step 2 – Undo division ($\times 2$): $w = 2$

3 Distributive Property

✦ **THE RULE:** Any number (or variable) outside the parentheses is multiplied by EVERYTHING inside. Pay close attention to positive/negative signs!

Example: $2(2x + 3) = -6$

Step 1 – Distribute: $2(2x) + 2(3) = 4x + 6 = -6$

Step 2 – Undo addition (subtract 6): $4x = -12$

✓ Step 3 – Undo multiplication ($\div 4$): $x = -3$

ERROR ANALYSIS A student expands $-3(x - 4)$ as $-3x - 12$. What went wrong?

A negative times a negative is positive: $-3 \cdot (-4) = +12$. Correct: $-3x + 12$. Distribute the sign to EVERY term inside the parentheses.

4 Variables on Both Sides

✦ **STRATEGY:** Move variables from the RIGHT to the LEFT using inverse operations, then solve like a two-step equation.

Example: $5x + 3 = 2x + 12$

Step 1 – Move variables (subtract $2x$): $3x + 3 = 12$

Step 2 – Undo addition (subtract 3): $3x = 9$

✓ Step 3 – Undo multiplication ($\div 3$): $x = 3$

5 Multi-Step Equations — The 5-Step Process

✦ **THE 5 STEPS:** 1. Distribute $\rightarrow$ 2. Combine Like Terms $\rightarrow$ 3. Move Variables $\rightarrow$ 4. Inverse for $+$ or $-$ $\rightarrow$ 5. Inverse for $\times$ or $\div$

Example: $-2x + 1 = 8(x + 2) - 5x$

1. Distribute ($8 \cdot x$ and $8 \cdot 2$): $-2x + 1 = 8x + 16 - 5x$

2. Combine like terms ($8x - 5x$): $-2x + 1 = 3x + 16$

3. Move variables ($+2x$): $1 = 5x + 16$

4. Undo addition (-16): $-15 = 5x$

✓ 5. Undo multiplication ($\div 5$): $x = -3$

6 Special Cases — No Solution & Infinite Solutions

🔑 **NO SOLUTION:** Variables cancel and you get a FALSE statement (like $5 = 9$).

🔑 **INFINITE SOLUTIONS:** Variables cancel and you get a TRUE statement (like $6 = 6$).

Result	Meaning	Example
Variables stay; $x = \text{number}$	One Solution	$2x+1=7 \rightarrow x=3$
Variables cancel; FALSE remains	No Solution	$2x+1=2x+5 \rightarrow 1=5 \times$
Variables cancel; TRUE remains	Infinite Solutions	$2x+1=2x+1 \rightarrow 1=1 \checkmark$

No Solution — Solve: $3(x + 2) = 3x + 10$

Distribute & subtract $3x$: $3x + 6 = 3x + 10 \rightarrow 6 = 10 \rightarrow \text{FALSE! No Solution}$

Infinite Solutions — Solve: $2(x + 4) = 2x + 8$

Distribute & subtract $2x$: $2x + 8 = 2x + 8 \rightarrow 8 = 8 \rightarrow \text{TRUE! Infinite Solutions}$

Quick Check

MC Solving an equation gives $7 = 7$. The equation has...

✓ Variables cancelled and a TRUE statement remains $\rightarrow$ infinitely many solutions.

7 Linear Inequalities

Fill the meaning, circle, and shade on your printed copy, then click to check.

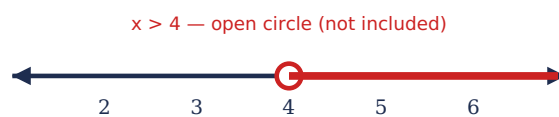
Symbol	Meaning	Circle	Shade
$<$	less than	OPEN	LEFT
$>$	greater than	OPEN	RIGHT
$\leq$	less than or equal	CLOSED	LEFT
$\geq$	greater than or equal	CLOSED	RIGHT

⚠ **CRITICAL RULE — FLIP THE SIGN:** When you MULTIPLY or DIVIDE both sides by a NEGATIVE number, you must FLIP the inequality sign! Example: $-2x > 6 \rightarrow$ divide by $-2 \rightarrow x < -3$ (sign flipped!)

Example (no flip): $2x + 3 > 11$

Subtract 3: $2x > 8$

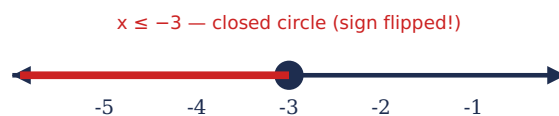
✓ Divide by 2 (positive – no flip): $x > 4$



Example (FLIP NEEDED): $-3x + 4 \geq 13$

Subtract 4: $-3x \geq 9$

✓ Divide by $-3 \rightarrow$ FLIP THE SIGN: $x \leq -3$



ERROR ANALYSIS To solve $-4x > 12$ a student wrote $x > -3$. What's the mistake?

Dividing both sides by the negative number -4 requires FLIPPING the sign. Correct: $x < -3$.

8 Word Problems — Equations & Inequalities

✦ **THE FLOWCHART:** 1) READ → 2) DEFINE the variable → 3) IDENTIFY keywords → 4) WRITE the equation/inequality → 5) SOLVE → 6) CHECK → 7) ANSWER in a sentence.

Equation (=)	Inequality	Addition	Subtraction
total, equals, is, the same as, gives, results in	at least ($\geq$), at most ($\leq$), more than ($>$), fewer than ($<$)	sum, plus, more than, increased by, total	difference, minus, less than, decreased by

SHORT ANSWER **Movie Theater:** A movie theater charges a \$6 entry fee plus \$4 per ticket. If the total cost was \$30, how many tickets were purchased?

Let x = number of tickets → $6 + 4x = 30$. Solve: $4x = 24$ → $x = 6$ → 6 tickets were purchased.

SHORT ANSWER **Notebooks:** A student wants to spend no more than \$50 on notebooks that cost \$6 each. How many notebooks can the student buy?

Let x = number of notebooks → $6x \leq 50$ ("no more than" = $\leq$). Solve: $x \leq 8.33$ → at most 8 notebooks.

[Go to Unit 3 Practice →](#)

Part 2 — Practice Problems

✦ **Show all steps on your printed copy.** Circle your final answer, then check on screen. Type just the value, e.g. for $x = 5$ type **5**.

A. Multi-Step Equations

1) $3x + 7 = 22$

answer

2) $2(x - 4) = 10$

answer

3) $5x - 3 = 2x + 9$

answer

4) $-4(x + 2) = 12$

answer

5) $6x + 4 = 4x - 8$

answer

6) $3(2x + 1) = 4x + 7$

answer

7) $x/4 + 5 = 8$

answer

8) $7 - 2x = 3x + 17$

answer

9) $0.5x + 3 = 7$

answer

10) $-(x - 6) = 2x + 3$

answer

B. Special Cases — No Solution or Infinite Solutions?

1) $2x + 3 = 2x + 3$

answer

2) $4x + 1 = 4x - 5$

answer

3) $3(x + 2) = 3x + 6$

answer

4) $5x - 5 = 5(x + 1)$

answer

5) $7x + 2 = 7x + 2$

answer

6) $x + 4 = x - 1$

answer

C. Inequalities — Solve (type like $x > 3$ or $x \leq 6$)

1) $3x + 2 > 11$

answer

2) $2x - 5 \leq 7$

answer

3) $-4x > 12$ (flip!)

answer

4) $5x + 1 \geq 16$

answer

5) $7 - x < 4$ (flip!)

answer

6) $2(x + 3) \leq 10$

answer

7) $-3x + 9 \geq 0$ (flip!)

answer

8) $4x - 1 > 2x + 5$

answer

D. Word Problems

1. A phone plan costs \$35/month plus \$0.10 per text. Last month's bill was \$52. How many texts were sent?

Let t = texts $\rightarrow 35 + 0.10t = 52$. Solve: $0.10t = 17 \rightarrow t = 170$ texts.

2. Sam has \$240 and saves \$30/week. Jake has \$420 and spends \$10/week. When will they have the same amount?

Let w = weeks $\rightarrow 240 + 30w = 420 - 10w$. Solve: $40w = 180 \rightarrow w = 4.5$ weeks.

3. A rectangle's perimeter is 56 cm. The length is 4 cm more than the width. Find the width.

Let w = width $\rightarrow 2(w + 4) + 2w = 56$. Solve: $4w + 8 = 56 \rightarrow w = 12$ cm.

4. Movie tickets cost \$9 each. You have at most \$50 to spend. What is the maximum number of tickets?

Let t = tickets $\rightarrow 9t \leq 50$ ("at most" = $\leq$). Solve: $t \leq 5.56 \rightarrow$ max 5 tickets (must be a whole number).

5. A car rental costs \$45/day plus \$0.25/mile. Your budget is \$120 for one day. How many miles can you drive?

Let m = miles $\rightarrow 45 + 0.25m \leq 120$. Solve: $0.25m \leq 75 \rightarrow m \leq 300$ miles.

6. Three consecutive even integers sum to 78. Find the smallest integer.

Let n = smallest $\rightarrow n + (n+2) + (n+4) = 78$. Solve: $3n + 6 = 78 \rightarrow n = 24$.

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